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The Production Relocation and Price Effects of US Trade Policy: The Case of Washing Machines

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1 Big picture

- **Question.** How do import restrictions affect consumer prices, foreign production locations, domestic production, and complementary goods in a modern global supply-chain setting?
- **Empirical setting.** The paper studies US washing machine trade policy: 2012 antidumping duties on Korea and Mexico, 2016 antidumping duties on China, and the 2018 Section 201 global safeguard tariffs on nearly all washer imports.
- **Why washing machines?** Washers are final consumer goods with detailed retail prices, easily tracked HS codes, a clear complement in dryers, and a concentrated set of multinational producers.
- **Trade-flow finding.** Country-specific antidumping duties induced production relocation: Korea/Mexico to China after 2012, then China to Thailand/Vietnam after 2016. The global 2018 safeguard tariffs instead led to front-running of imports and movement toward US production.
- **Retail-price finding.** The 2016 China antidumping duties generated only modest price changes. The 2018 safeguard tariffs raised washer prices by about 11–12 percent.
- **Complementary-good finding.** Dryer prices rose by a similar amount after the 2018 washer tariffs, even though dryers were not subject to new tariffs. This is central for measuring consumer incidence.
- **Domestic-brand finding.** Domestic brands raised prices along with foreign brands, consistent with strategic complementarities in oligopoly pricing.
- **Consumer incidence.** Including dryer price increases and domestic-brand price increases, the tariff elasticity of consumer prices for the 2018 safeguard tariffs is above one under the authors' preferred calculations.
- **Welfare and jobs.** The 2018 safeguard tariffs increased consumer costs by about \$1.55 billion annually. Net of tariff revenue, the implied annual consumer cost per job created was roughly \$817,000.
- **Economic takeaway.** The effect of tariffs depends crucially on whether production can relocate, whether policies are bilateral or global, and whether firms can spread price changes across complements and domestic competitors.

2 Incremental contribution of the paper

- **From import prices to retail prices.** The paper moves beyond border-price pass-through and measures final consumer prices using high-frequency retail microdata with product characteristics, brands, model age, and retailers.
- **From tariffs to firm relocation.** It documents how multinational producers respond to trade policy by shifting production across export platforms, not only by changing prices.
- **From one product to product systems.** It shows that tariffs on one product can raise prices of a complementary untaxed good, which changes how tariff incidence should be measured.
- **From foreign exporters to domestic competitors.** It highlights that domestic producers can raise prices when tariffs reduce import competition or increase rivals' costs.
- **From a single event to policy comparison.** By comparing bilateral antidumping duties and a near-global safeguard tariff, it shows why a tariff's geographic scope matters for both relocation and pass-through.
- **Contribution to trade-policy incidence.** The paper provides clean evidence on final-consumer tariff incidence in a large developed economy for a final durable good, where retail prices can be tracked precisely.
- **Contribution to multinational production.** It offers a rare event-study style view of export-platform relocation in response to trade barriers, showing that firms used plants in China, Thailand, Vietnam, and then the United States to avoid or respond to tariffs.

- **Contribution to antitrust/IO and pricing.** The dryer result connects trade policy to product complementarity, price bundling, and strategic pricing across product lines.

3 Data and measurement

3.1 Trade policy events

- **2012 antidumping duties.** Whirlpool petitioned against South Korean and Mexican washing machine imports. Duties were country- and firm-specific.
- **2016 antidumping duties.** New duties targeted washing machines from China.
- **2018 Section 201 safeguard tariffs.** Tariffs applied nearly globally: 20 percent on the first 1.2 million imported units per year and 50 percent thereafter. Parts were also covered by a tariff-rate quota.
- **Key contrast.** Antidumping cases allowed firms to relocate to third countries and keep serving the US market. The safeguard tariff covered nearly all source countries, making relocation to another foreign export platform less effective.

Interpretation. The sequence of policies creates a natural comparison between bilateral/country-specific trade barriers and broad global safeguards.

3.2 Trade and production-flow data

- US imports of washing machines and washing machine parts come from US International Trade Commission data.
- Korean exports of washing machine parts come from Korea Customs Service.
- Canadian imports and Korean imports are used to show third-country spillovers.
- PIERS bill-of-lading data allow the authors to identify firm-level shipment patterns for LG and Samsung.
- The core trade-flow variables are country of production, quantities imported, value imported, and parts shipments.

Interpretation. The paper uses trade-flow data to show that tariff responses are real production-location changes rather than merely accounting or statistical artifacts.

3.3 Retail price data

- Retail price data come from Gap Intelligence, a market research firm that records weekly prices and product characteristics.
- The sample covers major appliances: washers, dryers, ranges, refrigerators, and dishwashers.
- The analysis focuses on five major washer/dryer brands: LG, Samsung, Whirlpool, Maytag, and G.E.
- The retail sample focuses on five national retailers: JC Penney, Best Buy, Lowes, Sears, and Home Depot.
- Product characteristics include model, brand, first appearance, capacity, load type, Energy Star status, smart appliance status, and other product features.
- The authors use the net price after promotions and discounts rather than the sticker price.

Interpretation. High-frequency retail data let the authors measure consumer incidence directly and control for product life-cycle pricing and changes in product mix.

3.4 Control products

- Ranges are the main control category because they are similar to washers and dryers in steel content and import share.
- Refrigerators and dishwashers serve as placebo categories in the event-study figures and tables.
- The control categories absorb broader appliance-market shocks such as changes in steel and aluminum costs.

Interpretation. The empirical strategy compares washers and dryers to related durable goods, rather than simply comparing prices before and after tariffs.

3.5 Complementarity between washers and dryers

- Washers and dryers are often sold as matched pairs.
- Roughly three-quarters of washers have a matched dryer model.
- In more than 85 percent of weekly observations, matched washer-dryer models have exactly the same dollar price.
- Price levels and price changes of matched washer and dryer models are highly correlated.

Interpretation. This complementarity is why dryer prices matter for tariff incidence even though dryers were not directly taxed.

4 Models and empirical specifications

4.1 Hedonic price regression

The first price specification filters out price changes due to product mix, product features, and product life-cycle effects:

$$p_{irt} = \lambda_{C(i)t} + X_i\beta + b_{B(i)C(i)} + \sum_{a=2}^{25} \alpha_a^{C(i)} \mathbf{1}\{age_{it} = a\} + \gamma_r + \varepsilon_{irt}. \quad (1)$$

- i indexes product model, r retailer, and t week.
- $C(i)$ is the appliance category, such as washer or dryer.
- $B(i)$ is the brand.
- X_i contains product specifications.
- $\lambda_{C(i)t}$ is the category-week fixed effect.
- γ_r is a retailer fixed effect.

Interpretation. The category-week fixed effects trace residualized price movements for washers and dryers after holding product composition and life-cycle effects fixed.

4.2 Difference-in-differences event specification

To estimate tariff effects relative to control appliances, the paper estimates:

$$p_{irt} = \lambda_{C(i)t}^d + X_i\beta + b_{B(i)C(i)} + \sum_{a=2}^{25} \alpha_a^{C(i)} \mathbf{1}\{age_{it} = a\} + \gamma_r + \ell_t + \varepsilon_{irt}. \quad (2)$$

Here ℓ_t is a week fixed effect common to all appliances, while $\lambda_{C(i)t}^d$ measures category-week movements relative to ranges.

Interpretation. The week fixed effect removes common appliance shocks, including steel/aluminum cost shocks and broad retail-price changes. The category-week interaction isolates how washers and dryers move relative to controls.

4.3 Four-month and eight-month price effects

The four-month effect is calculated as:

$$\Delta_{event}^{4m} \bar{p}_C = (\bar{\lambda}_{C,-28:-20}^d - \bar{\lambda}_{C,-8:0}^d) - (\bar{\lambda}_{C,-8:0}^d - \bar{\lambda}_{C,+12:+20}^d). \quad (3)$$

An analogous eight-month effect uses windows $-44 : -36$ and $+28 : +36$.

Interpretation. The formula differences out pre-event trends. It asks whether the post-event relative price change exceeds the change implied by the pre-event trajectory.

4.4 Brand-specific estimates

The paper estimates analogous event effects separately by brand:

$$\Delta_{event}^h \bar{p}_{C,b} = f(\lambda_{C,b,t}^d), \quad h \in \{4m, 8m\}, \quad (4)$$

where b is brand.

Interpretation. Brand-specific estimates show whether price increases are concentrated among importers subject to tariffs or shared by domestic and foreign brands.

4.5 Tariff elasticity of consumer prices

The paper computes a tariff elasticity of consumer prices by dividing the estimated consumer price change by the relevant import-share-weighted tariff change:

$$\widehat{TECP}_{C,t} = \frac{\widehat{\Delta}_t^{8m} \bar{p}_C}{\left[\frac{M_{C,t-1}}{D_{C,t-1}} \right] \sum_{k \in K_C} s_{C,t-1}^k \Delta_t \tau_C^k - \left[\frac{M_{J,t-1}}{D_{J,t-1}} \right] \sum_{k \in K_J} s_{J,t-1}^k \Delta_t \tau_J^k}. \quad (5)$$

- M/D is the import share of demand.
- s^k is the pre-period trade share of source country or firm k .
- $\Delta \tau^k$ is the statutory tariff change.
- J denotes the control product category.

Interpretation. The numerator is the estimated consumer price change. The denominator is the effective tariff shock faced by the product category after weighting by import exposure and source shares.

5 Identification and empirical design

5.1 What identifies the price effects

- The treatment events are sharp tariff policy changes in 2016 and 2018.
- Treated products are washers, and indirectly dryers through complementarity.
- Control products are ranges, refrigerators, and dishwashers.
- Product characteristics and product age fixed effects absorb changes in product mix.
- Retailer fixed effects absorb persistent retail-chain pricing differences.
- Week fixed effects absorb common shocks affecting all appliances.
- Event windows remove pre-trends before calculating policy effects.

Interpretation. The design is a product-category difference-in-differences/event-study framework at weekly frequency.

5.2 Why the 2018 safeguard case is informative

- The safeguard tariff was broad enough to limit the country-hopping that occurred after the earlier anti-dumping duties.
- The timing of the tariff is clear, with implementation in February 2018.
- Washer prices rose after the tariff while placebo categories did not show comparable increases.
- Dryer prices also rose, revealing cross-product pricing spillovers.

Interpretation. The 2018 episode is the cleanest case for measuring direct consumer price pass-through because firms could not as easily route production through another foreign source country.

5.3 Why the 2012 and 2016 antidumping episodes matter

- They show that country-specific tariffs can be evaded or softened through relocation.
- Import quantities moved from Korea/Mexico to China, then from China to Thailand/Vietnam.
- Retail price effects were small or even negative because the tariff base shifted away from the targeted countries.

Interpretation. These episodes demonstrate that the incidence of trade policy depends on the production network response, not only on the statutory tariff rate.

5.4 Potential concerns and responses

- **Product mix.** Controlled with product characteristics, model fixed effects, and age controls.
- **Common cost shocks.** Controlled with week fixed effects and related appliance controls.

- **Seasonality.** Similar seasonality across appliance categories makes relative estimates less likely to be driven by seasonality.
- **Steel/aluminum tariffs.** Ranges help absorb metal-input cost shocks because they are similarly steel intensive.
- **Tariff anticipation.** Import front-running is documented directly, but price estimates use event windows and controls.
- **Missing product-level quantities.** The paper uses shipment/market-share data to corroborate retail-price conclusions and to calculate aggregate consumer costs.

Interpretation. The strongest evidence comes from the alignment of trade flows, firm-level sourcing, retail price movements, placebo products, and brand-specific price patterns.

6 Tables and figures

6.1 Figure 1: Trade Flows of Washing Machines and Washing Machine Parts + Figure 2: Firm-Level Imports of Washing Machines

Read:

- **Figure 1, Panel A** shows US washer imports shifting across source countries after each tariff action: Korea/Mexico to China, then China to Thailand/Vietnam, then front-running and decline around the 2018 safeguard.
- **Figure 1, Panel B** shows Korean exports of washer parts rising to the new production locations, consistent with supply-chain relocation rather than pure transshipment.
- **Figure 1, Panels C and D** show spillovers to Canada and Korea, indicating that US policy reshaped third-country trade flows.
- **Figure 2** uses PIERS shipment data and shows that LG and Samsung themselves shifted imports across Korea, China, Vietnam, and Thailand.
- The synchronized movement in parts shipments and finished washer imports supports the interpretation that firms moved production platforms.

Economic message. Country-specific tariffs changed geography more than total access to the US market. Multinational firms avoided duties by reallocating production across existing export platforms.

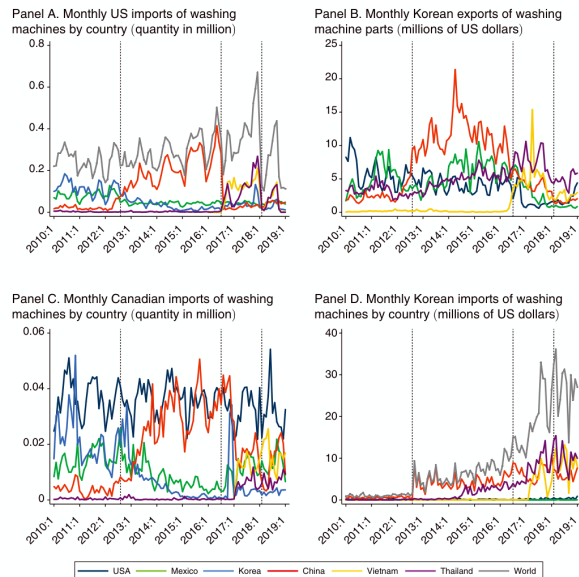


FIGURE 1. TRADE FLOWS OF WASHING MACHINES AND WASHING MACHINE PARTS

(a) Figure 1. Trade Flows of Washing Machines and Washing Machine Parts

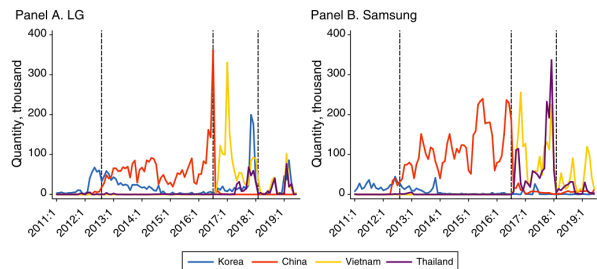


FIGURE 2. FIRM-LEVEL IMPORTS OF WASHING MACHINES

Note: See online Appendix C.14 for more details.

(b) Figure 2. Firm-Level Imports of Washing Machines

6.2 Figure 3: Average Unit Values by Production Location + Figure 4: Time Fixed Effects from Log Price Regression

Read:

- **Figure 3** shows that unit values fell after the 2012 Korea/Mexico antidumping duties when sourcing shifted to lower-priced China.
- The later China-to-Vietnam/Thailand shift does not produce a similarly visible decline in import unit values.
- Import unit values alone do not reveal final consumer incidence because they miss domestic-brand price responses and complementary-good pricing.
- **Figure 4** plots residualized washer and dryer prices from the hedonic regression and compares them to the CPI for laundry equipment.
- Washer and dryer prices both jump after the 2018 safeguard tariff, despite dryers not being taxed.
- The CPI and the retail microdata display broadly similar movements, validating the retail price series.

Economic message. Figure 3 shows why tariff incidence cannot be measured only at the border. Figure 4 shows that final retail incidence includes both washer and dryer pricing responses.

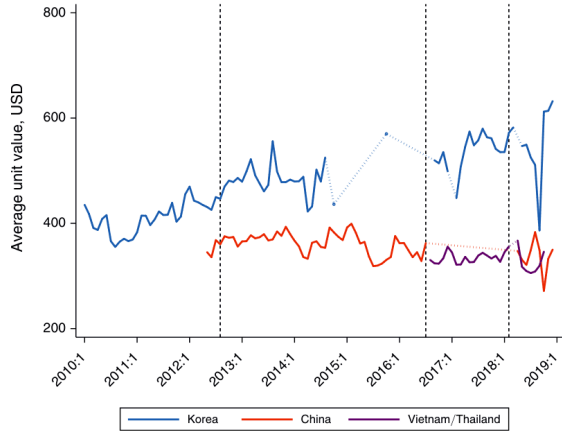


FIGURE 3. AVERAGE UNIT VALUES (EXCLUSIVE OF TARIFFS AND DUTIES) BY PRODUCTION LOCATION

includes residential washing machines classified under HS8450200040, HS8450200080, and HS8450

(a) Figure 3. Average Unit Values by Production Location

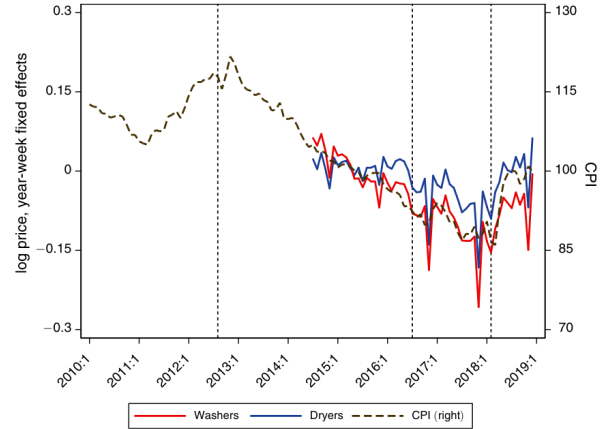


FIGURE 4. TIME FIXED EFFECTS FROM LOG PRICE REGRESSION, CPI FOR LAUNDRY EQUIPMENT

We plot the year-week fixed effects for washers or dryers, averaged over each month, obtained from

(b) Figure 4. Time Fixed Effects from Log Price Regression, CPI for Laundry Equipment

6.3 Figure 5: Price Effects of Safeguard Tariffs and Antidumping Duties against China + Table 1: Difference-in-Difference Estimates

Read:

- **Figure 5, Panel A** shows modest price effects around the 2016 China antidumping duties.
- **Figure 5, Panel B** shows a large post-2018 increase for washers and dryers relative to ranges, while refrigerators and dishwashers remain close to zero.
- The placebo categories strengthen the interpretation that the 2018 effect is specific to laundry products rather than a common appliance shock.
- **Table 1** quantifies the event-study patterns.
- After the 2016 China antidumping duties, washer prices rise only 2.6–5.8 percent depending on horizon/specification, and some placebo estimates are similar in columns 3–4.
- After the 2018 safeguard tariffs, washers rise 10.9–11.9 percent and dryers rise 11.1–11.9 percent.
- Refrigerators and dishwashers show near-zero or negative changes after the 2018 safeguard, supporting the identification strategy.

Economic message. The global safeguard tariff passed through strongly to final retail prices, while the China-specific antidumping duties had modest effects because production relocation remained possible.

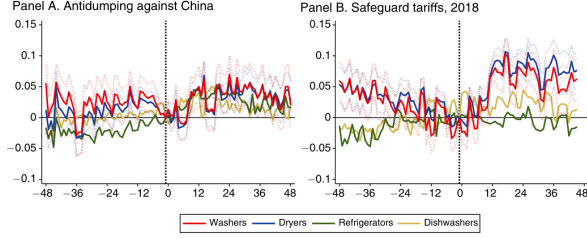


FIGURE 5. PRICE EFFECTS OF SAFEGUARD TARIFFS AND ANTIDUMPING DUTIES AGAINST CHINA

Notes: These figures report the regression coefficients $\lambda_{C(t)jt}$ from equation (2). In panel A the estimates are relative to the week of July 17, 2016, and in panel B the estimates are relative to the week of January 28, 2018. The dotted lines denote 95 percent confidence intervals for the coefficient estimates for washers and dryers, based on standard

(a) Figure 5. Price Effects of Safeguard Tariffs and Antidumping Duties against China

TABLE 1—DIFFERENCE-IN-DIFFERENCE ESTIMATES: PRICE EFFECTS OF WASHING MACHINE TARIFFS

	Antidumping against China				Safeguard tariffs 2018			
	4-month (1)	8-month (2)	4-month (3)	8-month (4)	4-month (5)	8-month (6)	4-month (7)	8-month (8)
Washers	0.026 (0.015)	0.034 (0.017)	0.046 (0.012)	0.058 (0.013)	0.109 (0.014)	0.115 (0.018)	0.110 (0.011)	0.119 (0.012)
Dryers	0.016 (0.012)	0.023 (0.014)	0.033 (0.009)	0.047 (0.010)	0.111 (0.013)	0.114 (0.017)	0.112 (0.009)	0.119 (0.009)
Refrigerators	0.025 (0.010)	0.008 (0.013)	0.039 (0.007)	0.028 (0.007)	0.001 (0.010)	-0.035 (0.015)	-0.002 (0.006)	-0.018 (0.007)
Dishwashers	0.012 (0.013)	-0.006 (0.014)	0.035 (0.008)	0.024 (0.008)	-0.010 (0.012)	-0.021 (0.018)	-0.012 (0.007)	-0.017 (0.009)
Model characteristics	✓	✓			✓	✓		
Model FE			✓	✓			✓	✓
Observations	1,637,298		1,637,298		1,637,298		1,637,298	

Notes: The table reports estimates for $\Delta_{\text{Domestic}}^{\text{Domestic}} \bar{p}_{C,t}$ and $\Delta_{\text{Foreign}}^{\text{Foreign}} \bar{p}_{C,t}$ defined in equation (3) and the text below it. The right hand side of equation (3) is a linear combination of the estimates from equation (2). Standard errors in parentheses.

(b) Table 1. Difference-in-Difference Estimates: Price Effects of Washing Machine Tariffs

6.4 Table 2: Brand-Specific Price Effects + Table 3: Tariff Elasticity of Consumer Prices

Read:

- **Table 2** shows that after the 2018 safeguard tariffs, all major brands raise washer and dryer prices.
- Domestic brands Whirlpool, Maytag, and G.E. raise prices even though they are not paying tariffs on imported finished washers in the same way as LG and Samsung.
- Whirlpool raises washer prices by about 13–17 percent; Maytag by about 14–17 percent; LG by about 8–13 percent; Samsung by about 15–18 percent; G.E. by about 5–7 percent.
- The broad brand response is consistent with strategic complementarities and reduced competitive pressure, not only mechanical cost pass-through by importers.
- **Table 3** decomposes consumer price changes, tariff-rate changes, and the resulting tariff elasticity.
- The 2012 Korea/Mexico antidumping episode has negative estimated price effects when washers and dryers are included, consistent with relocation to China lowering prices.
- The 2016 China antidumping episode implies a modest elasticity around 0.21–0.31.
- The 2018 safeguard tariffs imply much larger elasticities: 1.25 for washers only and 2.25 for washers plus dryers using the trade-weighted average tariff-rate change; even using the maximum tariff rate, the washer-plus-dryer elasticity is about 1.08.

Economic message. Table 2 shows that tariffs can raise prices even for domestic firms through market-structure responses. Table 3 shows why accounting for complements pushes consumer incidence above one.

TABLE 2—DIFFERENCE-IN-DIFFERENCE ESTIMATES: BRAND-SPECIFIC PRICE EFFECTS OF WASHING MACHINE TARIFFS

	Washers		Dryers		Refrigerators		Dishwashers	
	4-month	8-month	4-month	8-month	4-month	8-month	4-month	8-month
<i>Antidumping against China</i>								
Whirlpool	−0.031 (0.028)	−0.034 (0.036)	−0.034 (0.025)	−0.003 (0.034)	0.007 (0.017)	−0.012 (0.024)	−0.030 (0.021)	−0.046 (0.028)
Maytag	0.142 (0.044)	0.190 (0.053)	0.114 (0.037)	0.148 (0.046)	0.031 (0.049)	−0.014 (0.053)	0.012 (0.043)	−0.011 (0.048)
LG	0.010 (0.039)	−0.011 (0.049)	−0.008 (0.029)	−0.036 (0.042)	0.042 (0.028)	0.036 (0.042)	0.012 (0.058)	−0.024 (0.069)
Samsung	0.083 (0.041)	0.057 (0.050)	0.093 (0.036)	0.060 (0.041)	0.116 (0.037)	0.030 (0.044)	0.026 (0.061)	−0.054 (0.070)
G.E.	−0.002 (0.018)	0.024 (0.024)	−0.017 (0.017)	−0.018 (0.023)	−0.015 (0.016)	−0.008 (0.022)	0.012 (0.015)	0.011 (0.018)
<i>Safeguard tariffs, 2018</i>								
Whirlpool	0.174 (0.033)	0.129 (0.037)	0.175 (0.028)	0.142 (0.033)	0.008 (0.023)	−0.029 (0.029)	0.041 (0.024)	−0.001 (0.032)
Maytag	0.146 (0.035)	0.137 (0.050)	0.169 (0.031)	0.201 (0.047)	0.030 (0.028)	0.148 (0.048)	0.018 (0.029)	0.009 (0.058)
LG	0.081 (0.022)	0.131 (0.031)	0.082 (0.020)	0.125 (0.028)	0.040 (0.021)	0.022 (0.030)	0.136 (0.036)	0.158 (0.072)
Samsung	0.153 (0.031)	0.175 (0.039)	0.104 (0.028)	0.099 (0.035)	0.008 (0.021)	−0.071 (0.028)	−0.016 (0.024)	0.054 (0.050)
G.E.	0.072 (0.023)	0.051 (0.031)	0.123 (0.023)	0.108 (0.030)	−0.035 (0.017)	−0.064 (0.026)	−0.052 (0.017)	−0.066 (0.029)

Notes: The table reports results analogous to Table 1, based on separate estimates for each brand. Specifically, first equation (4) is estimated (with model characteristics as controls) and then a linear combination of these estimates is used to compute the left hand side of equation (3), separately for each brand and product category. Figure D3 in the online Appendix displays the corresponding weekly price estimates by brand. Standard errors are in parentheses.

TABLE 3—CALCULATION OF THE TARIFF ELASTICITY OF CONSUMER PRICES

	Antidumping		Safeguard Tariffs ^a	
	Korea and Mexico (1)	China (2)	Trade-weighted average (3)	Maximum (4)
<i>Panel A. Average consumer price change (%)^b</i>				
Washers only	N/A	3.40	11.50	11.50
Washers and dryers	−8.55	2.85	11.45	11.45
<i>Panel B. Pre-period trade-weighted average statutory tariff rate change (%)</i>				
Washers only	9.57	16.32	9.18	19.94
Washers and dryers	5.27	9.15	5.09	10.64
<i>Panel C. Tariff elasticity of consumer prices</i>				
Washers only	N/A	0.21	1.25	0.58
Washers and dryers	−1.62	0.31	2.25	1.08

Notes: Panels A and B follow the numerator and denominator of equation (6), respectively. The denominator calculates import shares in a pre-period defined as July–December 2011, July–December 2015, and February–August 2017 for the Korea/Mexico antidumping, China antidumping, and safeguard tariffs, respectively. Statutory tariff rate changes for antidumping duties use firm-level weights from PIERS Bill of Lading data (see online Appendix Table B2).

^a The two columns pertaining to the Section 201 safeguard tariffs correspond to the method of accounting for the heterogeneity in tariff rates over time. The *Trade-weighted average* uses import shares as weights during the period of study, and the *Maximum* column uses the maximum (50 percent) rate.

^b Columns 2, 3, and 4 in panel A are calculated based on estimates presented in column 2 of Table 1, where the estimates for washers and dryers are the simple average of the two coefficients. Column 1 is calculated using changes in the CPI for “laundry equipment” relative to “other appliances” during the relevant period (following equation (3)).

(a) Table 2. Brand-Specific Price Effects of Washing Machine Tariffs

(b) Table 3. Calculation of the Tariff Elasticity of Consumer Prices

7 Short summary

- The paper studies the effects of US trade policy on washing machines using trade data, firm-level shipping data, and weekly retail microdata.
- Country-specific antidumping duties led to production relocation across export platforms rather than large retail price increases.
- The 2018 global safeguard tariff limited country-hopping and produced a large retail price increase.
- Washer prices rose about 11–12 percent after the 2018 safeguard tariffs.
- Dryer prices rose by a similar amount despite not being subject to new tariffs.
- Domestic brands raised prices as well, consistent with strategic complementarities.
- The tariff elasticity of consumer prices is above one when dryer prices and domestic-brand price increases are included.
- The annual consumer cost of the safeguard tariffs is about \$1.55 billion, and the annual consumer cost per job is about \$817,000 net of tariff revenue.
- The policy lessons are that tariff incidence depends on global production networks, complementary goods, and market structure.

8 Conclusion

8.1 Core conclusion

Flaen, Hortaçsu, and Tintelnot show that the consumer price effects of trade policy depend on more than statutory tariff rates. The same industry experienced very different outcomes under bilateral antidumping duties and global safeguard tariffs. When firms could shift production to untaxed export platforms, price effects were small or even negative. When the tariff applied broadly, retail prices rose substantially.

8.2 Incremental contribution to the literature

The paper’s incremental contribution is to connect three literatures that are often separate. First, it contributes to tariff-incidence research by measuring final consumer prices instead of import prices. Second, it contributes to multinational-production and export-platform FDI research by documenting rapid relocation of production in response to country-specific tariff policy. Third, it contributes to industrial organization by showing that tariff incidence can spill over to complementary products and domestic competitors through firms’ pricing strategies.

8.3 Mechanism

The mechanism has three parts. First, country-specific tariffs create incentives to relocate production to third countries. Second, global tariffs make this avoidance strategy harder, which increases pass-through into retail prices. Third, firms in a concentrated product market may raise prices on both treated goods and complements. The dryer response is especially important because it reveals that firms did not simply pass through washer-specific costs; they reoptimized prices over the whole laundry-equipment product system.

8.4 Policy interpretation

The results imply that protectionist trade policy can have high consumer costs relative to the number of jobs created. They also show that a tariff’s incidence cannot be evaluated from tariffed products alone. If firms change prices of complements or if domestic firms raise prices in response to reduced import competition, the consumer cost of the tariff can exceed the direct tariff burden.

8.5 External validity and limitations

The washing machine industry is unusually useful for studying tariffs because products are final consumer durables with detailed retail prices and visible firm-level sourcing changes. But the same features mean that the exact estimates need not generalize mechanically to all industries. Relocation may be easier when multinationals already operate plants in alternative countries. Complementary-good pass-through may be stronger where products are commonly sold as matched pairs. Domestic price responses depend on industry concentration and substitution patterns.

8.6 Takeaway

The paper’s main lesson is that trade policy works through global production networks and product-market pricing systems. A tariff does not simply raise the cost of an imported good by a statutory percentage. It changes sourcing, product-line pricing, domestic competitors’ markups, and consumer costs across related goods.